

# Yilang Liu

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🌐 [Website](#) | 🐙 [GitHub](#) | 🔗 [LinkedIn](#) | 📄 [Google Scholar](#)

## SUMMARY

Yale Robotics Ph.D. candidate developing synthetic data pipelines, VLA policies, and sim-to-real learning methods. Deployed  $\pi_{0.5}$  in **18/20 successful factory trials** and transferred visual policies zero-shot to real robots. Publications: NeurIPS 2025 (**Spotlight**), CoRL 2026, RSS 2026, and ICRA 2026.

## EDUCATION

|                                                                                  |                                   |
|----------------------------------------------------------------------------------|-----------------------------------|
| <b>Yale University</b>                                                           | New Haven, CT                     |
| <i>Ph.D. Candidate in Mechanical Engineering (Robotics &amp; Robot Learning)</i> | <i>Expected May 2027</i>          |
| <b>Carnegie Mellon University</b>                                                | Pittsburgh, PA                    |
| <i>M.S. in Mechanical Engineering (Research Program)</i>                         | <i>May 2022</i>                   |
| <b>University of Cincinnati &amp; Chongqing University</b>                       | Cincinnati, OH / Chongqing, China |
| <i>B.S. in Mechanical Engineering (cum laude)</i>                                | <i>Apr 2020</i>                   |
| <i>B.E. in Mechanical Design, Manufacture and Automation</i>                     |                                   |

## INDUSTRY EXPERIENCE

|                                                                              |                            |
|------------------------------------------------------------------------------|----------------------------|
| <b>Dexmate Inc.</b> ( <i>bimanual mobile-manipulation robotics startup</i> ) | Santa Clara, CA            |
| <i>Robotics Intern</i>                                                       | <i>Oct 2025 – Jan 2026</i> |

- Fine-tuned and deployed Physical Intelligence’s  $\pi_{0.5}$  (pi0.5) VLA for bimanual condenser placement onto a conveyor, achieving **18/20 successful on-site factory trials**; stabilized execution at **~20 Hz** with Real-Time Action Chunking (RTC).
- Post-trained NVIDIA’s GR00T N1.6 VLA on [Dexmate Vega](#) for a long-horizon (**2-minute**) bimanual PVC-box unfolding-and-placement sequence, reaching **8/10 successful on-robot trials** after the zero-shot model failed.
- Collected and curated ~80 teleoperated demonstrations in LeRobot format; built the on-robot evaluation protocol and the two-GPU distributed fine-tuning pipeline (torchrun).
- Trained and deployed an ACT policy for vision-based bimanual bottle grasping and placement; developed an RRT-guided PPO policy for 6-DoF dexterous grasping in Genesis and Isaac Sim.

## SELECTED RESEARCH PROJECTS

|                                                     |                           |
|-----------------------------------------------------|---------------------------|
| <b>Graduate Research Assistant, Yale University</b> | New Haven, CT             |
| <i>CoCuRo Lab; Advisor: Prof. Ian Abraham</i>       | <i>Aug 2022 – Present</i> |

|                                                          |                                                                          |
|----------------------------------------------------------|--------------------------------------------------------------------------|
| <b>EmbodiedSWE: coding agents for dexterous robotics</b> | ICLR 2027 — under review                                                 |
| Co-first author / project co-lead                        | <a href="#">[paper]</a> <a href="#">[website]</a> <a href="#">[code]</a> |

- Co-developed a robotics benchmark spanning **28 scenes and 5 embodiments**, with contact-rich manipulation, deformable objects, and task executions lasting up to **30 minutes**.
- Built a trajectory-generation pipeline with scene and strategy diversification, domain randomization, and replay verification; fine-tuned **SmolVLA across 8 simulation tasks** using **10–400 generated demonstrations per task**.
- Reconstructed the lab using **3D Gaussian splatting** and generated synthetic robot demonstrations in a simulation environment incorporating the reconstruction.
- Fine-tuned  $\pi_{0.5}$  on the generated demonstrations and achieved **zero-shot sim-to-real deployment** on a Franka arm with a Robotiq gripper for grasping, unscrewing, and removing a light bulb.

## Sampling-Guided Policy Search (SGPS)

ICRA 2027 — under review

First author [\[paper\]](#)

- Developed SGPS, combining recurring sampling-based MPC action refinement with first-order policy gradients to learn visual locomotion and manipulation policies on a **single RTX 4080**.
- Distilled specialists into one visual policy and transferred it **zero-shot to a real Go2** for autonomous trotting, crawling, hurdle traversal, and behavior switching using onboard depth, without online MPC.

## Efficient visual reinforcement learning (SDPG)

CoRL 2026

Co-first author [\[paper\]](#) [\[code\]](#) [\[website\]](#)

- Co-led algorithm design and implemented the open-source codebase; estimates visual-policy gradients from perturbed rollouts without differentiating through rendering, using **about 1/5 of PPO's GPU memory**.
- Built visual RL benchmarks and led hardware experiments: trained a domain-randomized Go2 policy in **2 hours** and deployed it **zero-shot** on stairs, boxes, and uneven terrain.

## Sampling-based hybrid mode control for agile quadruped skills

ICRA 2026

First author [\[paper\]](#) [\[website\]](#)

- Designed hybrid mode-sequence optimization with asymptotic-optimality guarantees; completed all three agile skills (foot stand, jump flip, handstand), with **~39% lower cost** than the best fixed-sequence baseline.
- Deployed **50 Hz** predictive control on a laptop-class CPU with EKF state estimation, achieving standing-to-handstand transitions on a real Go2 without motion capture.

## Graduate Student Researcher, Carnegie Mellon University

Pittsburgh, PA

Advisor: Prof. Amir Barati Farimani

Aug 2020 – May 2022

- Developed an energy-saving snake-robot locomotion controller with an online model-based policy-gradient RL method in physics simulation; published in ASME J. Mechanisms and Robotics (2023).

## TECHNICAL SKILLS

**Programming & ML:** Python, C++, PyTorch, JAX, Hugging Face Transformers; VLA fine-tuning ( $\pi_{0.5}$ , GR00T N1.6, SmolVLA), imitation learning (ACT, behavior cloning), visual RL, PPO.

**Robotics & Control:** ROS/ROS 2, MPC, motion planning, EKF, teleoperation, sim-to-real transfer; Franka, Robotiq, Unitree A1/Go2/G1.

**Simulation & Infrastructure:** MuJoCo/MJX, Isaac Sim/Lab/Gym, Genesis, Brax, 3D Gaussian splatting; LeRobot, Docker, Linux, Git, Weights & Biases, distributed training (torchrun).

## SELECTED PUBLICATIONS & MANUSCRIPTS

Y. Liu\*, Z. Shen\*, H. You\*, et al. “[EmbodiedSWE: Coding Agents for Long-Horizon Dexterous Robotics](#)”. CoRL Workshop on Agentic Robotics (\*equal contribution and project leadership). A subsequent version is **under review at ICLR 2027**.

Y. Liu, H. You, Q. Wang, D. Rakita, I. Abraham. “[Accelerating Visual Policy Learning with Sampling-Based Model Predictive Control](#)”. **ICRA 2027, under review**.

H. You\*, Y. Liu\*, D. Zong, et al. “[Efficient On-policy Visual-RL via Stochastic Decoupled Policy Gradient](#)”. CoRL 2026 (\*equal contribution).

Y. Liu, H. You, I. Abraham. “[Sample-Based Hybrid Mode Control: Asymptotically Optimal Switching of Algorithmic and Non-Differentiable Control Modes](#)”. ICRA 2026.

H. You, Y. Liu, I. Abraham. “[Accelerating Visual-Policy Learning through Parallel Differentiable Simulation](#)”. NeurIPS 2025 (**Spotlight**).

C. Hughes, Y. Liu, et al. “[Asymptotically Optimal Ergodic Coverage on Generalized Motion Fields](#)”. RSS 2026.

Full list: [Google Scholar](#).

## TEACHING & SERVICE

**Teaching:** Yale ENAS 773, Robotics, Control, and Learning (TF); CMU 18-461, Machine Learning (TA).

**Reviewer:** ICRA (2022–2026), IROS (2022–2026), RSS (2025), ECCV (2024), IEEE RA-L.